

MANAV NARULA

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Professional Summary

AI Engineer with 1.9+ years of experience building AI-powered systems using Generative AI, Computer Vision, and backend APIs. Experienced in LLM pipelines, RAG architectures, AI Agents and multimodal data extraction, delivering scalable solutions for real-world business workflows.

Work Experience

Billtech Labs

Sep 2024 - Present

AI Engineer

- Developed a multimodal AI pipeline to extract structured data from 300+ jewellery CAD images daily using **GPT-4o** (OCR + semantic parsing), generating JSON with **17+ attributes** and automated Excel outputs via **Pandas**
- Built a **FastAPI** backend integrated with **Meta WhatsApp Webhooks** to handle image/document ingestion, perform format conversion, and asynchronously deliver processed outputs (images + text) to users.
- Engineered **Flask**-based REST APIs to integrate with external processing systems using file-based pipelines (inbound/outbound), enabling asynchronous request handling and automated JSON responses.
- Developed data management APIs for pricing, labour, and rate configurations using Excel and file-based storage with validation and error handling.
- Built an OCR pipeline using **EasyOCR** and **OpenCV** to extract text with bounding boxes and confidence scores, generating structured JSON for downstream processing.
- Designed an **LLM**-based pipeline to convert unstructured text into structured JSON and automated Excel generation using **Pandas**.
- Integrated **Gemini Vision API** for automated background removal and enhancement of jewellery images using prompt-based image transformations.

Key Projects

Hybrid RAG + AI Agent System | *Python, LangChain, OpenAI API, FAISS, Pandas Agent* **GitHub** | **Live Demo**

- Built an AI application using **LangChain** and **OpenAI** to route queries between **FAISS (RAG)** and **Pandas DataFrame agent** based on query intent.
- Implemented multi-format ingestion (PDF, TXT, CSV) with LLM-based document classification for invoices, contracts, and bank statements.

Smart Document Inspector | *Python, FastAPI, OpenAI Vision API, Prompt Engineering*

GitHub

- Developed a **FastAPI backend** using OpenAI Vision models for document processing with structured JSON output and intelligent document classification.

Education

Guru Gobind Singh Indraprastha University

2021 - 2024

Bachelor of Computer Applications (BCA)

New Delhi

Technical Skills

- Generative AI & LLMs:** OpenAI API (GPT-4, GPT-4o), Google Gemini API, Prompt Engineering, Function Calling, Structured Outputs
- Languages:** Python, SQL
- Machine Learning:** Scikit-learn, Classification, Regression, Feature Engineering, Model Evaluation, ML Pipelines
- RAG & AI Agents:** LangChain, FAISS, Vector Databases, Pandas Agent, Query Routing, Embeddings
- Computer Vision & OCR:** OpenAI Vision API, EasyOCR, OpenCV, Document AI, Image Preprocessing
- Backend Development:** FastAPI, Flask, REST APIs, Webhooks, API Integration, Microservices
- Tools & Cloud:** AWS EC2, Git, Docker, Linux, Streamlit, Postman, WhatsApp Cloud API, Pandas, Excel Automation